

Name: _____

Class: _____

ACTIVITY SHEET

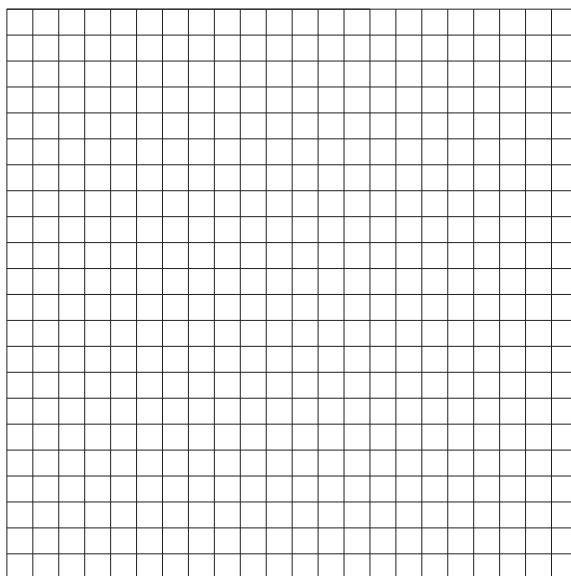
3.4 Graphing data and interpreting graphs

The table below shows the melting and boiling points of the first eight straight-chain alkanes.

Alkane	Formula	MP (°C)	BP (°C)
Methane	CH ₄	−183	−162
Ethane	C ₂ H ₆	−172	−88
Propane	C ₃ H ₈	−188	−42
Butane	C ₄ H ₁₀	−138	0
Pentane	C ₅ H ₁₂	−130	36
Hexane	C ₆ H ₁₄	−95	69
Heptane	C ₇ H ₁₆	−91	98
Octane	C ₈ H ₁₈	−57	126

Instructions

Plot the melting points and boiling points on the grid below and then connect the plotted points. Ensure the graph has a title, axis labels and a key. Add a horizontal dashed line to indicate room temperature (20°C).



Questions

- 1
 - a Which of the alkanes are gases at room temperature?
 - b How do you know?
- 2
 - a Which of the alkanes are liquids at room temperature?
 - b How do you know?
- 3
 - a Which of the alkanes are solids at room temperature?
 - b How do you know?
- 4 Describe the trends in the melting points of the alkanes.
- 5 Describe the trends in the boiling points of the alkanes.
- 6 What factor might be causing the increasing melting and boiling points?
- 7 Predict the melting point of nonane (C_9H_{20}).
- 8 Predict the boiling point of nonane (C_9H_{20}).